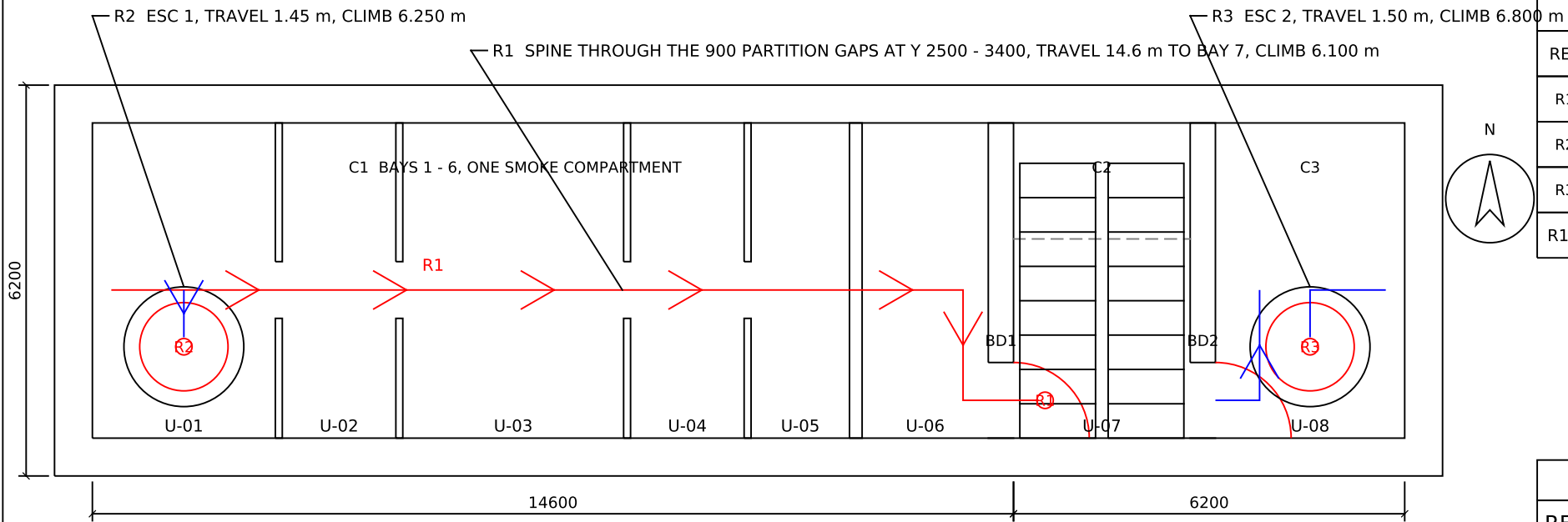
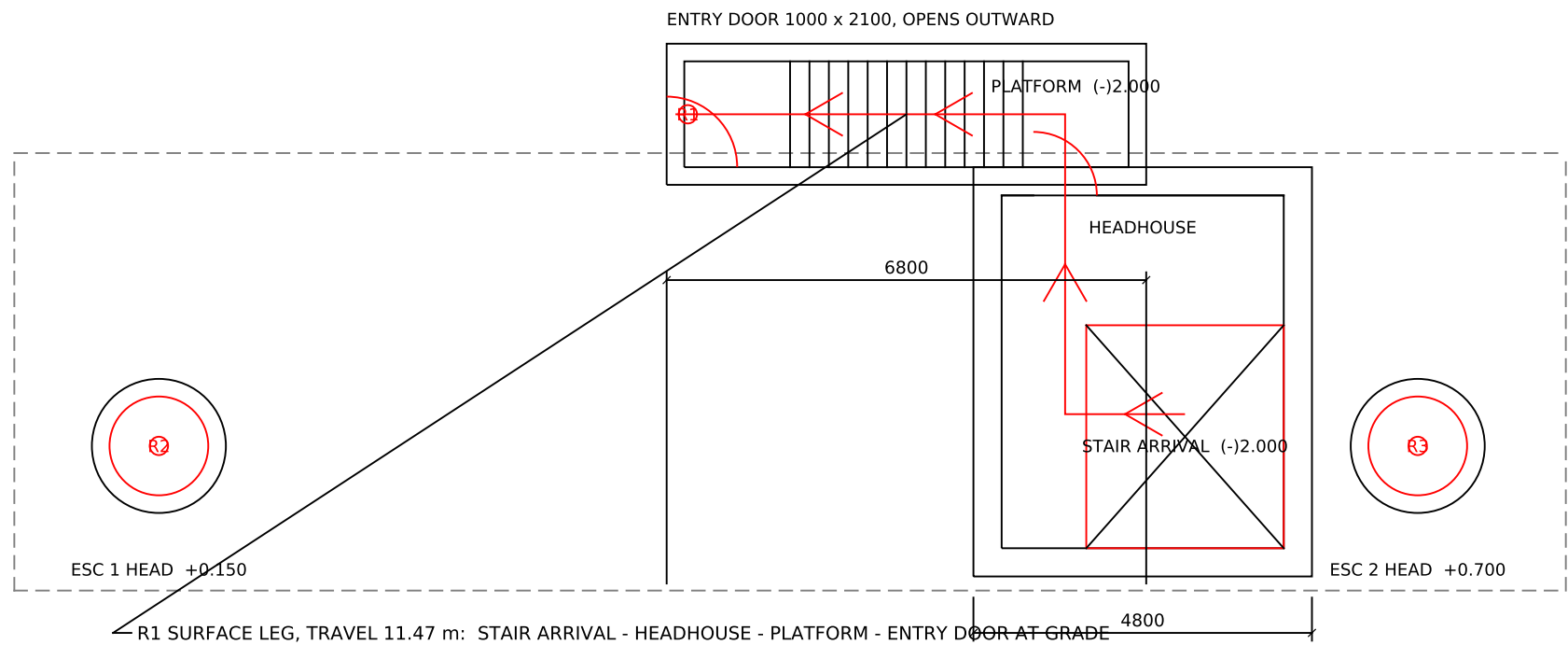
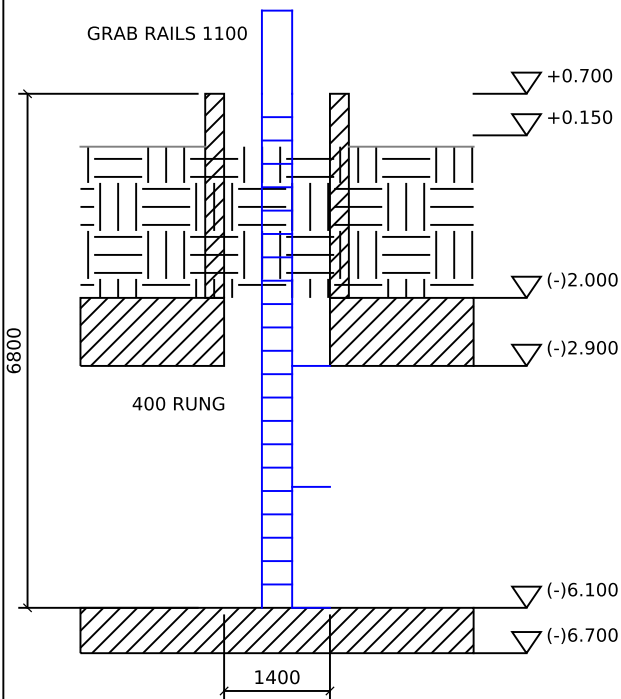




1 UNDERGROUND LEVEL ESCAPE PLAN (-)6.100
1 : 100 SPINE 20.800 m; THREE ROUTES, R1 PRIMARY



2 ENTRY LEVEL ESCAPE PLAN (-)2.000 AND 0.000
1 : 100 R1 EMERGES OUTSIDE THE PROTECTIVE BOUNDARY



3 ESCAPE SHAFT - SECTION
1 : 100 ESC 1 AND ESC 2

WHERE THE FIRE IS	ROUTE TO USE	AND WHAT IS SHUT
BAY 8, GENERATOR	R1; R2 IF THE SPINE IS SMOKE-LOGGED	BLAST DOOR 2 SHUT. DO NOT USE R3
BAYS 1 TO 6, OCCUPIED	R1 IF BAY 7 IS CLEAR	OTHERWISE R2 FROM THE WEST, R3 FROM THE EAST
BAY 7, STAIR SHAFT	R2 OR R3	BOTH BLAST DOORS SHUT. R1 IS THE FIRE
HEADHOUSE OR STAIRWELL	R2 OR R3	R1'S SURFACE END IS BLOCKED

4 EVACUATION DECISION RULE
NOT TO SCALE

ESCAPE ROUTE SCHEDULE				
REF	ROUTE	TRAVEL	CLIMB	WHERE IT GOES
R1	PRIMARY - MAIN STAIR	14.6 m	6.100 m	BAY 7 SHAFT, HEADHOUSE, THEN 12R TO THE ENTRY DOOR AT GRADE
R2	ESCAPE SHAFT ESC 1	1.45 m	6.250 m	BAY 1, (2050, 2050), 1400 DIA. HEAD +0.150
R3	ESCAPE SHAFT ESC 2	1.50 m	6.800 m	BAY 8, (19900, 2050), 1400 DIA. HEAD +0.700
R1 USES THE MAIN STAIRCASE: 24R AT 170.8333, TREAD 280, 3 FLIGHTS OF 8, TOTAL RISE 4100, WIDTH 1200, HEADROOM 2533				

POINT OF EMERGENCE SCHEDULE				
REF	EMERGES AT	OPENING	LEVEL	NOTES
R1	ENTRY DOOR, COVERED STAIRWELL	1000 x 2100	0.000	OPENS OUTWARD. OUTSIDE THE BOUNDARY
R2	ESC 1 HEAD, BAY 1	1400 DIA	+0.150	6.250 m CLIMB ON THE LADDER
R3	ESC 2 HEAD, BAY 8	1400 DIA	+0.700	6.800 m CLIMB ON THE LADDER

FIRE AND SMOKE COMPARTMENT SCHEDULE		
REF	COMPARTMENT	NOTES
C1	BAYS 1 - 6, THE OCCUPIED ZONE (U-01 TO U-06)	ONE SMOKE COMPARTMENT; THE FOUR W8 GAPS ARE PERMANENT
C2	BAY 7, THE STAIR SHAFT (U-07)	BOUNDED BY BLAST DOOR 1 IN W6 AND BLAST DOOR 2 IN W7
C3	BAY 8, GENERATOR AND ESC 2 (U-08)	THE LIKELIEST FIRE LOAD IN THE SHELTER; ESC 2 IS INSIDE IT
BLAST DOORS 1 AND 2 ARE THE ONLY REAL BARRIERS BETWEEN THE THREE COMPARTMENTS. W5 IS DESIGNATED FIRE AND GAS-TIGHT.		

ESCAPE SHAFT LADDER SCHEDULE			
ITEM	SPECIFICATION	EXTENT	NOTES
RUNGS	20 DIA GALVANISED MS	400 CLEAR WIDTH	EQUAL PITCH WITHIN EACH SHAFT
PITCH, ESC 1	297.6 mm	21 SPACES	6.250 m CLIMB
PITCH, ESC 2	295.7 mm	23 SPACES	6.800 m CLIMB
STRINGERS	2 No. 50 x 10 GALVANISED FLAT	-	CAST-IN LUGS TO THE 250 COLLAR AND THE ROOF-SLAB BORE
BRACKETS	EXPANSION ANCHORED	1.5 m CENTRES	OVER THE LOWER 3.200 m
CLEARANCES	200 OR MORE BEHIND THE RUNG	750 OR MORE CLIMBING SPACE IN FRONT	GRAB RAILS 1100 ABOVE THE HEAD

EVACUATION DECISION RULE		
WHERE THE FIRE IS	ROUTE TO USE	AND WHAT IS SHUT
BAY 8, GENERATOR	R1; R2 IF THE SPINE IS SMOKE-LOGGED	BLAST DOOR 2 SHUT. DO NOT USE R3
BAYS 1 TO 6, OCCUPIED	R1 IF BAY 7 IS CLEAR	OTHERWISE R2 FROM THE WEST, R3 FROM THE EAST
BAY 7, STAIR SHAFT	R2 OR R3	BOTH BLAST DOORS SHUT. R1 IS THE FIRE
HEADHOUSE OR STAIRWELL	R2 OR R3	R1'S SURFACE END IS BLOCKED

UNDERGROUND CBRN-HARDENED
BLAST-RESISTANT PROTECTIVE
STRUCTURE + SENTRY POST
PUNE, MAHARASHTRA

NOTES

- ALL DIMENSIONS ARE IN MM. ALL LEVELS ARE IN METRES RELATIVE TO FINISHED SITE GRADE 0.000, NEGATIVE DOWNWARDS.
- WRITTEN DIM TO BE FOLLOWED ONLY. DRG NOT TO BE SCALED.
- THIS DRG IS TO BE READ IN CONJUNCTION WITH ALL RELEVANT ARCH, STR AND SERVICES DRGS.
- THERE ARE THREE ESCAPE ROUTES AND NO OTHERS. R1 IS THE MAIN STAIR IN BAY 7 TO THE HEADHOUSE AND OUT THROUGH THE COVERED ENTRY STAIRWELL; R2 AND R3 ARE THE TWO 1400 DIA ESCAPE SHAFTS.
- BAYS 1 TO 6 ARE ONE SMOKE COMPARTMENT. THE FOUR W8 PARTITIONS ARE NON-STRUCTURAL AND CARRY PERMANENT 900 GAPS. SO BLAST DOORS 1 AND 2 IN W6 AND W7 ARE THE ONLY REAL BARRIERS IN THE SHELTER.
- ESC 2 IS IN BAY 8 WITH THE GENERATOR. FOR A FIRE IN BAY 8, BLAST DOOR 2 IS SHUT AND R3 IS NOT USED - EVACUATE BY R1, OR BY R2 IF THE SPINE IS SMOKE-LOGGED.
- EACH SHAFT IS CLIMBED ON A FIXED LADDER: 20 DIA GALVANISED MS RUNGS AT 400 CLEAR WIDTH, EQUAL PITCH, 200 OR MORE BEHIND THE RUNG AND 750 OR MORE CLIMBING SPACE IN FRONT. WITH GRAB RAILS 1100 ABOVE THE HEAD.
- THE MAIN STAIRCASE IS 24 RISERS AT 170.8333, TREAD 280, 3 FLIGHTS OF 8, TOTAL RISE 4100, WIDTH 1200, HEADROOM 2533.
- KEEP EVERY ROUTE, EVERY DOOR SWING AND BOTH SHAFT BASES CLEAR AT ALL TIMES.
- CONTRACTOR TO CHECK AND VERIFY THE DRG BEFORE EXECUTION.

FIRE AND LIFE SAFETY - ESCAPE PLAN, SHAFT SECTION AND DECISION RULE

CBRN HARDENED UG OPS ROOM

SHEET 12

Project number	01
Date	18 SEP 2026
Drawn by	SYN 01
Checked by	DR IR CHAUDHARI
FLS012	
Scale	As indicated